

Contestant ID: _____

Time: _____

Rank: _____



MANAGERIAL ACCOUNTING (135) STATE 2023

CONCEPT KNOWLEDGE:

Multiple Choice (20 @ 2 points each) _____ (40 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (75 points)

Problem 1: CVP Analysis _____ (30 points)

Problem 2: Profitability Analysis _____ (30 points)

Problem 3: Process Costing _____ (30 points)

TOTAL POINTS _____ **(205 points)**

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts: concept knowledge and application knowledge.

Your name and/or school name should *not* appear on any work you submit for grading. Write your Member ID in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

- Round all calculations to two decimal places at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

- Which of the following changes to a company's contribution income statement will always lower the break-even point (either in units or in dollars)?
 - Sales price increases by 10%.
 - Sales price decreases by 5%.
 - Variable costs increase by 10% and fixed costs decrease by 5%.
 - Variable costs decrease by 5% and fixed costs increase by 10%.
- Eaton Manufacturing currently produces 4,000 bicycles per month. The following per unit data apply for sales to regular customers:

Direct materials	\$53
Direct manufacturing labor	10
Variable manufacturing overhead	12
Fixed manufacturing overhead	14
Total manufacturing costs	89

The plant has capacity for 6,000 bicycles and is considering expanding production to 5,000 bicycles. What is the per unit cost of producing 5,000 bicycles?

- \$91.80 per unit.
 - \$111.25 per unit.
 - \$69.80 per unit.
 - \$86.20 per unit.
- Cost A is a fixed cost, while B is a variable cost. During the current year, the volume of production has decreased. In terms of cost per unit of production, we would expect that:
 - Cost A has remained unchanged.
 - Cost B has decreased.
 - Cost A has decreased.
 - Cost B has remained unchanged.

4. Given the following costs for Larry's Liquidators, select the answer choice that correctly classifies each cost's behavior.

Total Cost at:			
1,500 Units		1,800 Units	
Cost A	\$4,650	Cost A	\$5,580
Cost B	8,600	Cost B	9,400

- A. Cost A and Cost B are both variable
 B. Cost A is variable; Cost B is mixed
 C. Cost A is variable; Cost B is fixed
 D. Cost A and Cost B are both mixed
5. In the computation of the cost per equivalent unit, the weighted-average method of process costing considers all the costs:
- A. entering work in process from the units in beginning inventory plus the costs for the work completed during the current accounting period
 B. that have entered work in process from the units started or transferred in during the current accounting period
 C. that have entered work in process during the current accounting period from the units started or transferred in minus the costs associated with ending inventory
 D. that have entered work in process during the current accounting period from the units started or transferred in plus the costs associated with ending inventory
6. Which of the following is true of weighted-average process-costing?
- A. It does not represent the average cost of units when inputs prices fluctuates markedly from month to month.
 B. It facilitates period-to-period comparisons and hence is very useful in analyzing the performances of managers for different periods.
 C. It arrives at the same unit costs as arrived under FIFO method, but the computations are easier under weighted-average process-costing.
 D. It calculates the cost per equivalent unit of all work done to date, regardless of the accounting period in which it was done.

7. A golf shoe manufacturer makes and sells thousands of golf shoes each year through two channels: wholesale to sporting goods stores and direct to consumers through its custom portal. Customers can specify modifications to 6 basic golf shoe models including spiked or spikeless, traditional or casual, even laced or other closures. From there, customers can specify many details to their preferred model such as size and width (different for each foot), favorite colors for the base, saddle, accents, and laces. A customer can choose to embroider a name or initials or include the logo of their favorite MLB, NFL, NHL or NCAA team. Which of the following costing systems would make the most sense for the custom golf shoe operation and the main product operation (not customized/wholesale)?
 - A. operation-costing and job costing
 - B. job costing for both operations
 - C. operation-costing and process costing
 - D. process costing for both operations
8. Which of the following is a reason that has accelerated the demand for refinements to the costing system?
 - A. The declining demand for customized products has led managers to decrease the variety of products and services their companies offer.
 - B. The use of product and process technology has led to an increase in indirect costs and a decrease in direct costs.
 - C. The increase of automated processes has led to the increase in direct manufacturing cost leading to a decrease in break-even point.
 - D. The increasing competition in product markets has led to an increase in contribution margin resulting in a decrease of break-even point.
9. Which of the following causes Work-In-Process to increase (assume normal costing)?
 - A. Incurring factory utilities
 - B. Moving a completed job to finished goods
 - C. Using direct materials in production
 - D. Accruing the wages of indirect laborers
10. In process costing, a separate work in process account is kept for each:
 - A. Individual order
 - B. Equivalent unit
 - C. Processing department
 - D. Cost category (i.e., materials, conversion cost).

11. Which one of these best reflects a distinguishing factor between a job order cost system and a process cost system?
- A. The detail at which costs are recorded
 - B. The time period each covers
 - C. The number of work-in-process accounts
 - D. The manufacturing cost elements included
12. All of the following statements are correct when referring to process costing except:
- A. Process costing would be appropriate for a jeweler who makes custom jewelry to order.
 - B. A process costing system has the same basic purposes as a job-order costing system.
 - C. Units produced are indistinguishable from each other.
 - D. Costs are accumulated by department.
13. Which of the following is not an assumption of CVP analysis?
- A. All units produced are sold
 - B. All costs are variable
 - C. Sales mix remains constant
 - D. The behavior of costs and revenues and linear within the relevant range
14. At the break-even point of 2,000 units, variable costs are \$55,000 and fixed costs are \$32,000. How much is the selling price per unit?
- A. \$43.50
 - B. \$11.50
 - C. \$16.00
 - D. \$32.00
15. The following information is available for Robinson Company:

Sales	\$600,000	Total fixed expenses	\$150,000
Cost of goods sold	390,000	Total variable expenses	360,000

- A CVP income statement would report
- A. Gross profit of \$210,000
 - B. Contribution margin of \$450,000
 - C. Gross profit of \$240,000
 - D. Contribution margin of \$240,000

16. The break-even point is
- A. Total sales equal total variable costs
 - B. Contribution margin equals total fixed costs
 - C. Total variable costs equal total fixed costs
 - D. Total sales equal total fixed costs
17. Ramirez Company requires sales of \$2,000,000 to cover its fixed costs of \$700,000 and to earn net income of \$500,000. What percent are variable costs of sales?
- A. 25%
 - B. 40%
 - C. 35%
 - D. 60%
18. Which of the following costs at a manufacturing company would be treated as a product cost under variable costing?
- A. direct material cost
 - B. property taxes on the factory building
 - C. sales manager's salary
 - D. sales commissions
19. Net operating income computed under variable costing would exceed net operating income computed using absorption costing if:
- A. units sold exceed units produced.
 - B. units sold are less than units produced.
 - C. units sold equal units produced.
 - D. the average fixed cost per unit is zero.
20. Last year, Harold Company produced 1,500 units and sold 500 units. The company had no beginning inventory. Harold incurred the following costs:

Direct materials per unit	\$34
Direct labor per unit	\$12
Variable overhead per unit	\$14
Total fixed manufacturing overhead	\$6,000
Sales Price per unit	\$180

Contribution margin per unit under variable costing is _____.

- A. \$120
- B. \$116
- C. \$60
- D. \$64

Short Answer

- McGonagall Company sells only two products, Product A and Product B. McGonagall sells two units of Product A for each unit it sells of Product B.

	Product A	Product B	Total
Selling price	\$50	\$20	
Variable cost per unit	\$17	\$4	
Total fixed costs			\$2,132,000

How many units of Product A does McGonagall need to sell to break-even?	
How many units of Product B does McGonagall need to sell to break-even?	

- Potter Company has the following data for the Assembly Department for August:

Units in process at the beginning of August	700
Units started in August	2,200
Units completed and transferred	?
Units in process at end of August	800

Conversion costs are added evenly throughout the process. The company uses the weighted-average method. Compute the equivalent units of production for direct materials and conversion costs for each independent scenario:

	Direct Materials	Conversion Costs
Units in process at the end of August are 20% complete; materials are added at the beginning of the process.		
Units in process at the end of August are 80% complete; materials are added at the beginning of the process.		
Units in process at the end of August are 45% complete; materials are added at the end of the process.		
Units in process at the end of August are 78% complete; materials are added at the halfway point.		

3. HPU is an American beverage company that manufactures energy drinks. The company calculates product cost per pallet of finished and packaged product. For each pallet, \$3,100 of direct material is used and there is \$2,500 of direct manufacturing labor at \$25 per hour. Manufacturing overhead is applied at \$30 per direct manufacturing labor hour.

Calculate the profit earned on 45 pallets if each pallet sells for \$10,000.	
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4. Sirius, Inc. reports the following information for July:

Sales Revenue	\$1,000,000
Variable Costs	120,000
Operating Income	390,000

What is the total fixed cost using variable costing?	
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5. Carrey Company manufactures a single product. Annual production costs incurred in the manufacturing process are show below for the production of 2,000 units. The utilities and maintenance are mixed costs; the fixed portion of these costs are \$200 and \$400, respectively. Complete the chart below, determining the total costs for the production of 4,000 units

	Costs Incurred	
Production in Units:	2,000	4,000
Production Costs:		
Direct Materials	\$4,000	
Direct Labor	16,000	
Utilities	1,000	
Rent	3,000	
Indirect Labor	4,600	
Supervisor Salaries	1,500	
Maintenance	900	
Depreciation (straight-line)	2,500	

6. Daniel is considering opening a Kwik Oil Change Center. He estimates that the following costs will be incurred during his first year of operations: Rent \$6,000, Deprecation on equipment \$7,000, Wages \$16,400, Motor oil \$1.80 per quart. He estimates that each oil change will require 5 quarts of oil. Oil filters will cost \$3.00 each. Daniel must also pay The Kwik Corporation a franchise fee of \$1.40 per oil change since he will operate his business as a franchise. In addition, utility costs are expected to behave in relation to the number of oil changes as follows:

Number of Oil Changes	Utility Costs
4,000	\$6,000
6,000	7,300
19,000	15,000
12,000	12,600
9,400	10,200

Daniel anticipates that he can provide the oil change service with a filter for \$25.00 each.

Using the high-low method, determine variable cost per unit for utilities.	
Using the high-low method, determine fixed costs for utilities.	
Determine the break-even point in number of oil changes	
Determine the break-even point in sales dollars	
Determine the number of oil changes required to earn net income of \$20,000.	

Problem 1 (30 Points)

Little Whinging Company sells only two products, Product A and Product B.

	Product A	Product B	Total
Selling price	\$40	\$50	
Variable cost per unit	\$24	\$40	
Total fixed costs			\$840,000

Little Whinging sells two units of Product A for each unit it sells of Product B. Little Whinging faces a tax rate of 30%.

How many units of Product A does Little Whinging need to see to break-even?	
How many units of Product B does Little Whinging need to see to break-even?	
How many units of Product A would need to be sold if Little Whinging desired an after-tax net income of \$73,500?	
How many units of Product B would need to be sold if Little Whinging desired an after-tax net income of \$73,500?	
If Little Whinging's tax rate is reduced to 25%, how many units of Product A would need to be sold to earn after-tax net income of \$73,500?	
If Little Whinging's tax rate is reduced to 25%, how many units of Product B would need to be sold to earn after-tax net income of \$73,500?	

Problem 2 (30 Points)

Rubeus Co. manufactures and sells adjustable windows for remodeling homes and new housing. Rubeus developed its budget for the current year assuming that the windows would sell at a price of \$500 each. The variable costs for each window were forecasted to be \$250 and the annual fixed costs were forecasted to be \$130,000. Rubeus had targeted a profit of \$450,000.

While Rubeus's sales usually increase during the second quarter, the May financial statements reported that sales were not meeting expectations. For the first five months of the year, only 400 units had been sold at the established price, with variable cost as planned, and it was clear that the target profit for the year would not be reached unless some actions were taken. Rubeus's president assigned a management committee to analyze the situation and develop several alternative courses of action. The following three alternatives were presented to the president, only one of which can be selected.

- Alternative 1: Reduce the selling price by \$50. The marketing department forecasts that with the lower price, 3,200 units could be sold during the remainder of the year.
- Alternative 2: Lower variable costs per unit by \$30 through the use of less expensive materials. Because of the difference in materials, the selling price would have to be lowered by \$40 and sales of 2,600 units for the remainder of the year are forecast.
- Alternative 3: Cut fixed costs by \$15,000 and lower the selling price by 5 percent. Sales of 2,200 units would be expected for the remainder of the year.

If no changes are made to the selling price or cost structure, estimate the number of units that must be sold during the year to break-even.	
If no changes are made to the selling price or cost structure, estimate the number of units that must be sold during the year to attain the target profit of \$450,000.	
Calculate the profitability of each of the alternatives: • Alternative 1	
• Alternative 2	
• Alternative 3	
Which alternative should Rubeus choose?	

Problem 3 (30 Points)

R. Lupin Company was founded by Remus Lupin and David Thewlis in 1929. Early Lupin fishing rods were made from Bamboo, a classic style at the time. Eventually Lupin made the transition from bamboo to fiberglass to graphite. Today, Lupin builds state-of-the-art boron graphite fishing rods. The company uses a weighted-average process-costing system and has one production department for the production of their graphite rods.

All materials are introduced at the start of manufacturing; in contrast, conversion cost is incurred uniformly throughout production. The company had respective work-in-process inventories on May 1 and May 31 of 62,200 units and 71,000 rods, the latter of which was 50% complete. The production supervisor noted that 102,000 rods were completed during the month.

Costs in the May 1 work-in-process inventory were subdivided as follows: materials, \$41,000; conversion, \$92,000. During May, the company's cost accountant charged production with \$339,600 of material and \$730,000 of conversion.

Determine the number of rods that the company started during May.	
Compute the number of equivalent units with respect to conversion cost.	
Determine the conversion cost per equivalent unit.	
Compute the cost of the May 31 work-in-process inventory.	
Compute the cost of goods manufactured in May.	
What account would have been credited to record the company's completed production?	